CONCRETE BUILDING: OBJECTIVES & REQUIREMENTS

Project Assignment for ARC 2713 Structural Systems 1

REFERENCE FOR SCOPE OF ASSIGNMENT

See class Lecture 12 & CONCRETE BUILDING_Parameters & Prototypes for an overview of the project scope and parameters.
See also the video recording of the digital model ALPHA_V2 presented in class for reference to a range of possible horizontal and vertical structural element configurations and relationships.
Apply lessons learned from our field trip to the Southfield parking garage with a precast concrete structure.
See Precast Prototypes for your assigned prototype; either ALPHA, CHARLIE, DELTA, or KILO Posters on Canvas.
Note: Use color-coding of your building elements with the eight-color range from Lecture 12. Each distinct and typical structural element to have a unique color in your model.

OBJECTIVES

Demonstrate understanding of basic precast concrete structural elements & systems in response to loads, spans, element configuration, and element sizing.
Demonstrate understanding of basic modes of structural representation including graphical and mathematical communication.
Demonstrate ability to plan and configure the structure of a simple precast concrete building.
Demonstrate ability to apply structural principles and structural concepts of economy.

REPRESENTATION and SUBMISSION

A single pdf, 16x9 aspect ratio, landscape format, color, clear line and text resolution.
Organize content with the same general sheet format used for the two previous projects.
Upload to Canvas under Assignments. Each student submission will be evaluated using the Evaluation Rubric.
Due by 11:59 pm on Monday, November 16: A single consolidated pdf file (<10MB), meeting Objectives & Requirements.

CONCRETE BUILDING REQUIREMENTS & PROJECT FORMATTING GUIDELINES

TITLE SLIDE with PROJECT PARAMETERS and ASSIGNED PRECAST PROTOTYPE

Course name and Project name, your name and date of submission
List the Project Parameters including use, location, and roof loads and floor loads specific to use and location.
List the Project Parameters for your assigned Precast Prototype model (either ALPHA, CHARLIE, DELTA, or KILO).
Axon or Perspective of your assigned Precast Prototype model (from ALPHA, CHARLIE, DELTA, or KILO Poster).
Axon image of “Finite Elements” of the assigned Precast Prototype.

SLIDE: ROOF FRAMING PLAN DIAGRAM

Depict a Plan Diagram of the roof structure. Include grid lines and label with letters in one direction and numbers in the other direction. Include overall dimensions of the building, and dimensions between each grid line. Show calculation of the total roof load (Snow load and Dead load) required for your roof structure design.
On this diagram depict, label, and calculate for the following structural elements:

Roof Deck (Hollow-Core or Double-Tee) supporting roof load
Roof deck elements (Hollow-Core or Double-Tee) with typical element width and span dimension. Graphically diagram and calculate the Tributary Area in sf, list the Tributary Load in psf, and calculate the Total Tributary Load in plf delivered by a typical roof deck element to a typical beam or a vertical load-bearing panel.

Vertical Load-Bearing Panel (Lite-Wall or Solid) supporting roof deck or beam
Typical Panel indicating panel width & thickness. Graphically diagram and calculate the Tributary Area in sf, list the Tributary Load in psf, and calculate the Total Tributary Load in plf for one typical Vertical Load-Bearing Panel.

Beam supporting roof deck
Beams with span dimension and Beam type (R, T, or L). Graphically diagram and calculate the Tributary Area in sf, the Tributary load in psf, and the Total Tributary Load in plf for one typical roof Beam.

Column supporting roof beam
Graphically diagram and calculate the Tributary Area in sf, the Tributary load of the Beam in Kips (include self-weight of Beam), and the Total Tributary Load in Kips for one typical Column.

Shear Walls
Label the shear walls, identifying at least two that resist lateral load in the transverse direction and two in the longitudinal direction.

SLIDE: ROOF DECK ELEMENT & SIZING

Roof Deck (Hollow-Core or Double-Tee)
Depict the span & load table for your precast concrete roof deck element and highlight the allowable load and span closest to that required for your structure design.
Depict the section image of the precast concrete roof deck element with the cross-sectional dimensions.
List the self-weight of the element in psf and plf
List and directly compare the allowable load for the element to the actual load for your design.
List and directly compare the allowable span to the actual span for your design.

SLIDE: FLOOR FRAMING PLAN DIAGRAM

Depict a Plan Diagram of the 2nd-Floor structure. Include grid lines and label with letters and numbers (same as roof plan diagram). Include overall dimensions of the building, and dimensions between each grid line. Show calculation of the total floor load (Live load and Dead load) required for your floor structure design. On this diagram include the two structural stair elements, one floor opening for an elevator, and note the plan area which is open to the main floor level below.

On this diagram depict, label, and calculate for the following structural elements:

Floor Deck (Hollow-Core or Double-Tee) supporting roof load

Floor deck elements with typical element width and span dimension. Graphically diagram and calculate the Tributary Area in sf, list the Tributary Load in psf, and calculate the Total Tributary Load in plf delivered by a typical floor deck element to a typical beam or a vertical load-bearing panel.

Vertical Load-Bearing Panel (Lite-Wall or Solid) supporting floor deck or beam

Typical Panel indicating panel width & thickness. Graphically diagram and calculate the Tributary Area in sf, list the Tributary Load in psf, and calculate the Total Tributary Load in plf for one typical Vertical Load-Bearing Panel.

Beam supporting floor deck

Beams with span dimension and Beam type (R, T, or L). Graphically diagram and calculate the Tributary Area in sf, the Tributary load in psf, and the Total Tributary Load in plf for one typical Beam.

Column supporting floor beam

Graphically diagram and calculate the Tributary Area in sf, the Tributary load of the Beam in Kips (include self-weight of Beam), and the Total Tributary Load in Kips for one typical Column.

Shear Walls

Label the shear walls, identifying at least two that resist lateral load in the transverse direction and two in the longitudinal direction.

Stairs

Label the stairs with stair width and span.

SLIDE: FLOOR DECK ELEMENT & SIZING

Depict the span & load table for your precast concrete floor deck element and highlight the allowable load and span closest to that required for your structure design. Depict the section image of the element with the cross-sectional dimensions. List the self-weight of the element in psf and plf

List and directly compare the allowable load for the element to the actual load for your design.

List and directly compare the allowable span to the actual span for your design.

Depict the section image of the precast concrete floor deck element with the cross-sectional dimensions, and include

SLIDE: AXON OF ALL TRIBUTARY ELEMENTS TRANSFERRING LOAD TO ONE COLUMN AND FOUNDATION

Create an Exploded Assembly Axon drawing that depicts ***only*** all of the tributary elements and area (both roof and floor) transferring load to one typical Column and the column Foundation. This will include selected areas of roof & floor deck elements and beams.

Depict load arrows in the direction of the load paths for each structural element transferring load to the Column.

Depict a load arrow of the overall Column load transferred to the Foundation.

Calculate and list the Total Concentrated Point Load on the Foundation, which includes the self-weight of the Column.

Depict and highlight the span & load table for the roof Beam and floor Beam, indicating minimum size (width and depth) of each structural beam.

SLIDE: ONE TRANSVERSE SECTION DRAWING

Depict all structural elements for roof and floor structure, including structural Foundation as a Mat Slab supporting vertical structure. The Mat Slab should be either 50” thick (for Ypsilanti or Pinconning) or 68” thick (for Houghton).

Label and list all final structural element sizes and spans for floor and roof structure, and size and thickness of foundation

Include a line that indicates grade at and beyond the building. Indicate the Main Floor level (top of Mat Slab) at 8” above grade.

SLIDES: TWO AXONS (one Axon per slide)

Depict the final building structure with two Axon drawings, one showing two sides and roof, and the other showing the other two sides and roof. The drawings should have all structural elements color-coded. Include Foundation.

SLIDES: ASSEMBLY SEQUENCE

3D Axons showing assembly sequence of structural elements from floor to roof, including shear wall elements.

Depict in slide order: first a slide of the Mat Slab Foundation, then a slide of all full height Vertical Structure & Shear Walls, then a slide adding Floor Beams, then Stairs, then Floor Deck, then Roof Beams, the Roof Deck. You may include roof openings for skylights, just don’t compromise the structural integrity of the Roof Deck elements.

FINAL SLIDE: PERSPECTIVE & PROJECT SUMMARY

Perspective image of Concrete Building, at eye-level for a person standing at grade level, human figures for scale.

Written summary of how your design responds to structural concepts of economy, using a minimum of material and an optimized layout to achieve maximized structural work.